

SMART INDIA HACKATHON 2026



Problem Statement ID - 26175

Problem Statement Title - DepthWizard — Single-View Height Estimation and 3D Flythrough

Theme - Disaster Management

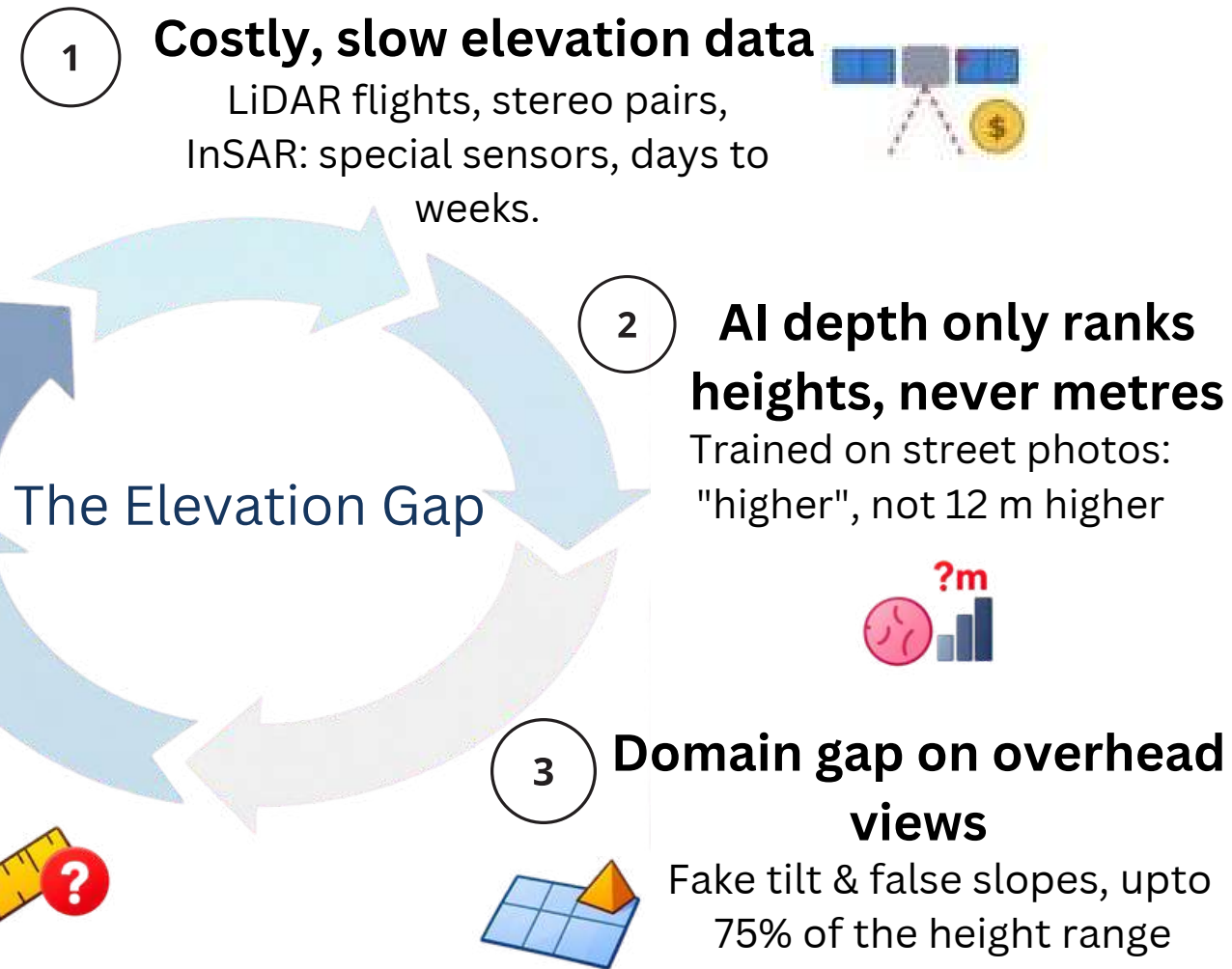
PS Category - Software

Team ID - 175991

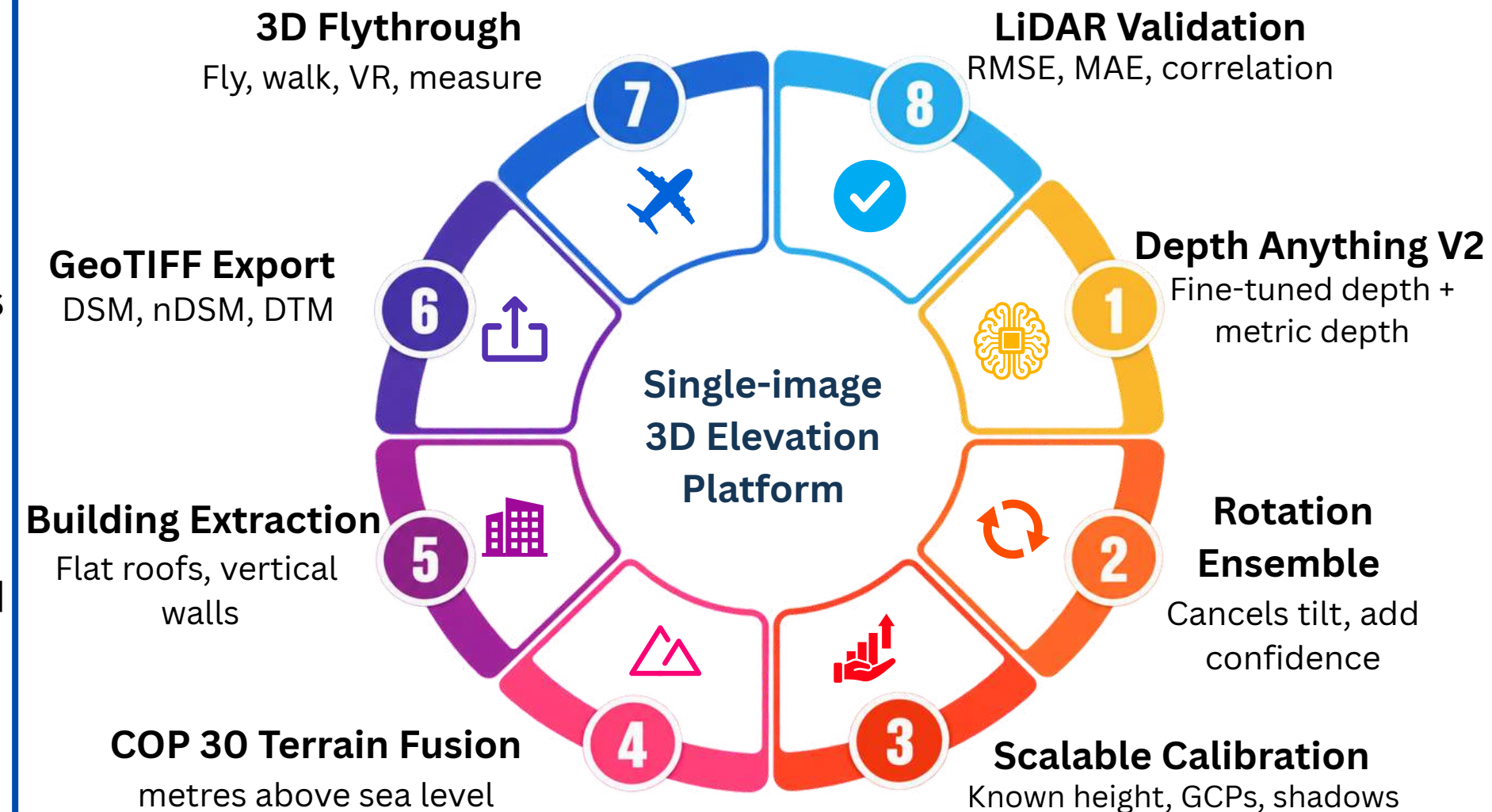
Team Name - Alcoholic1



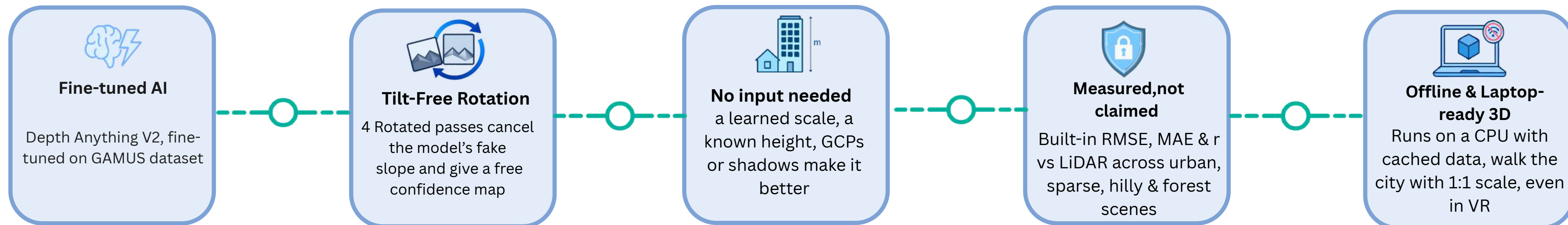
PROPOSED PROBLEM

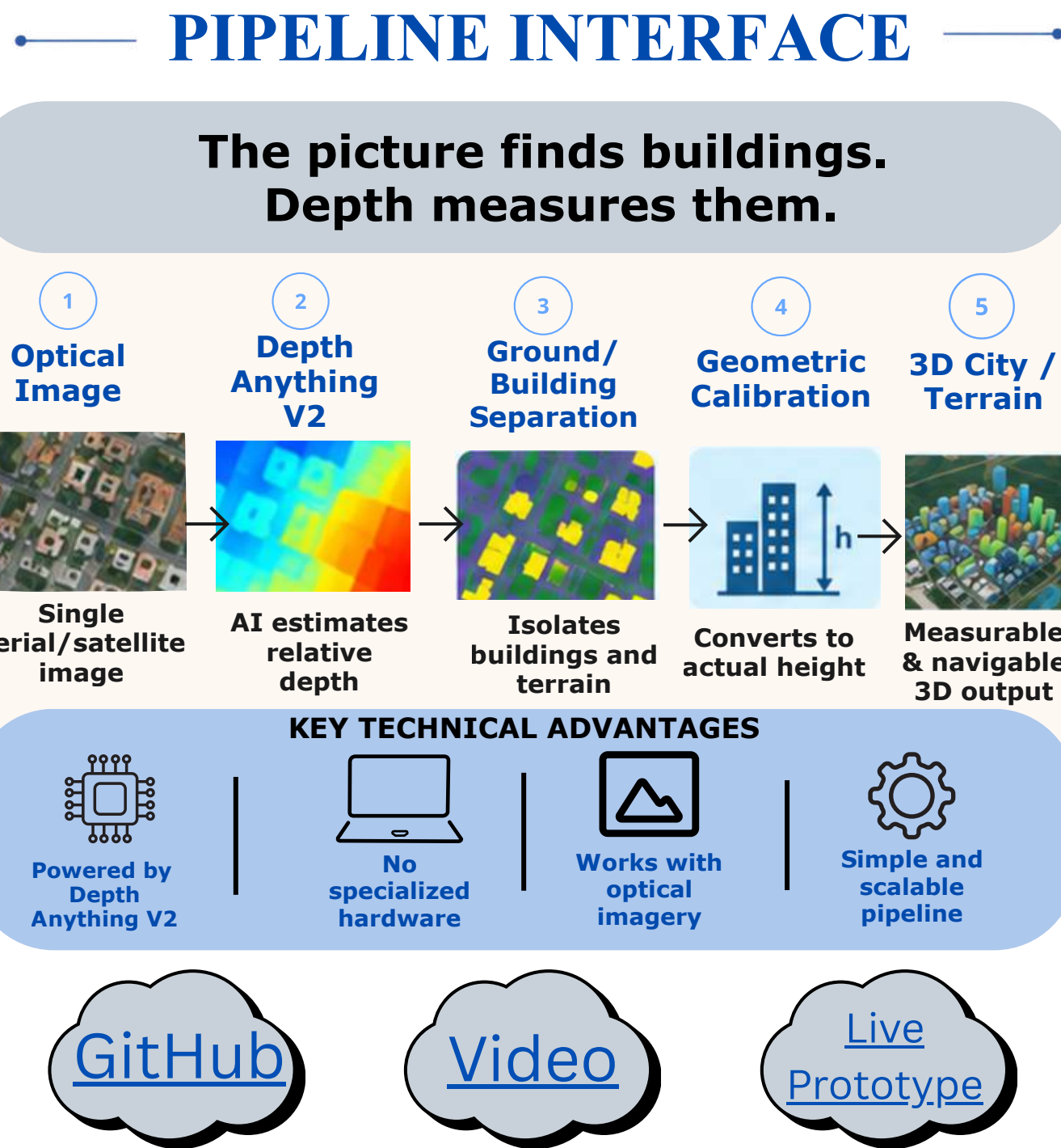
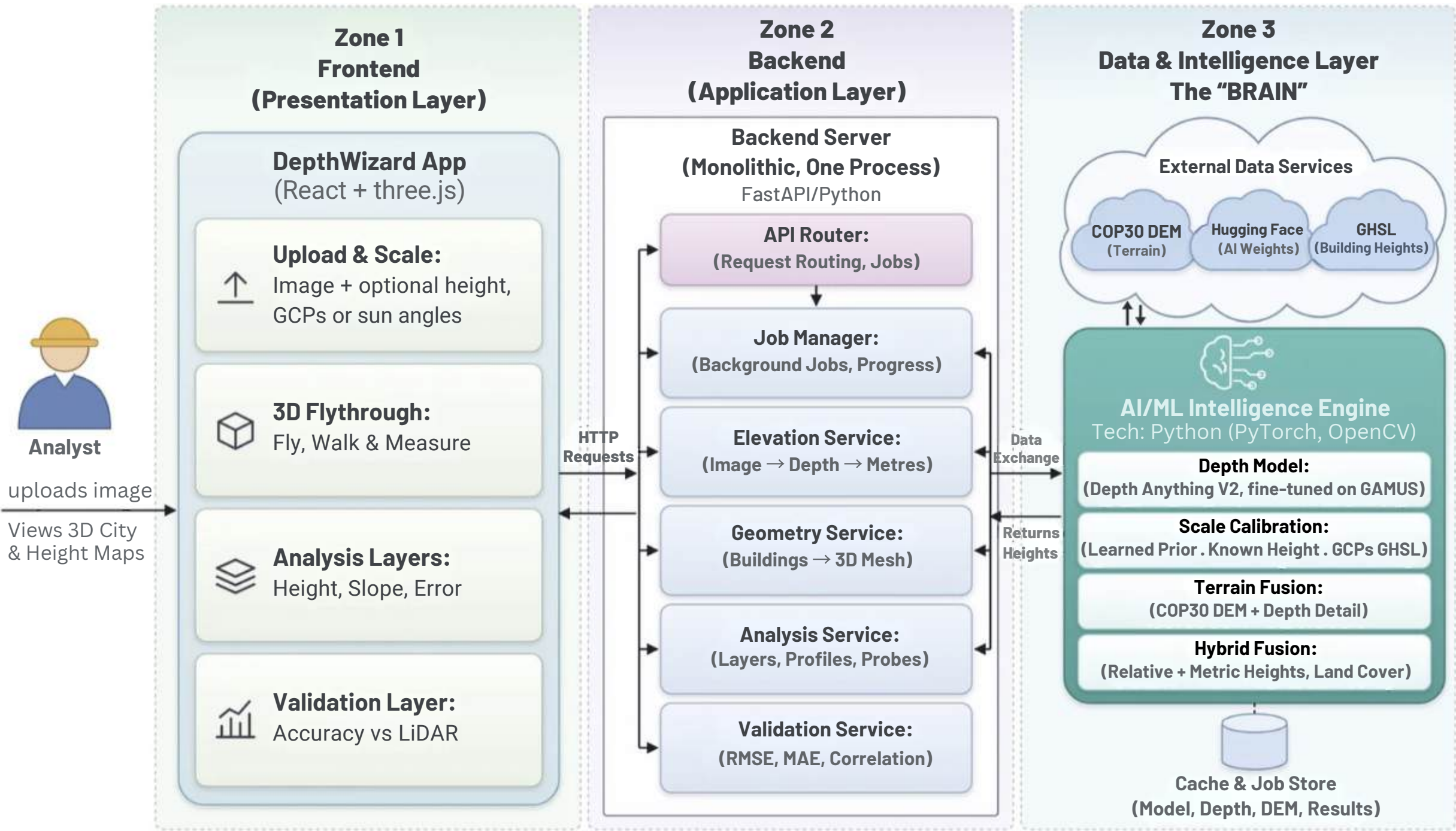


PROPOSED SOLUTION



Innovation & Uniqueness





FEASIBILITY

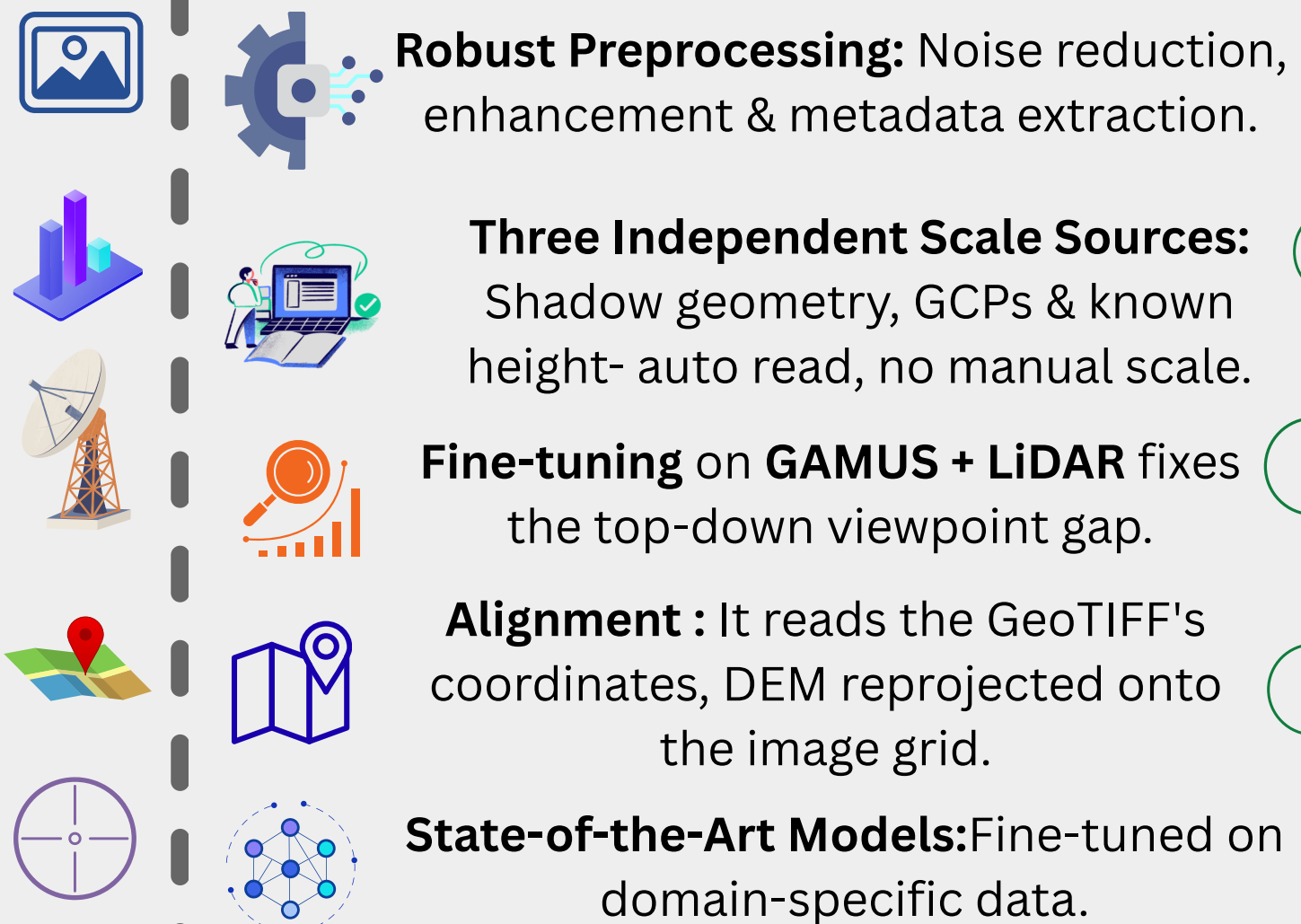
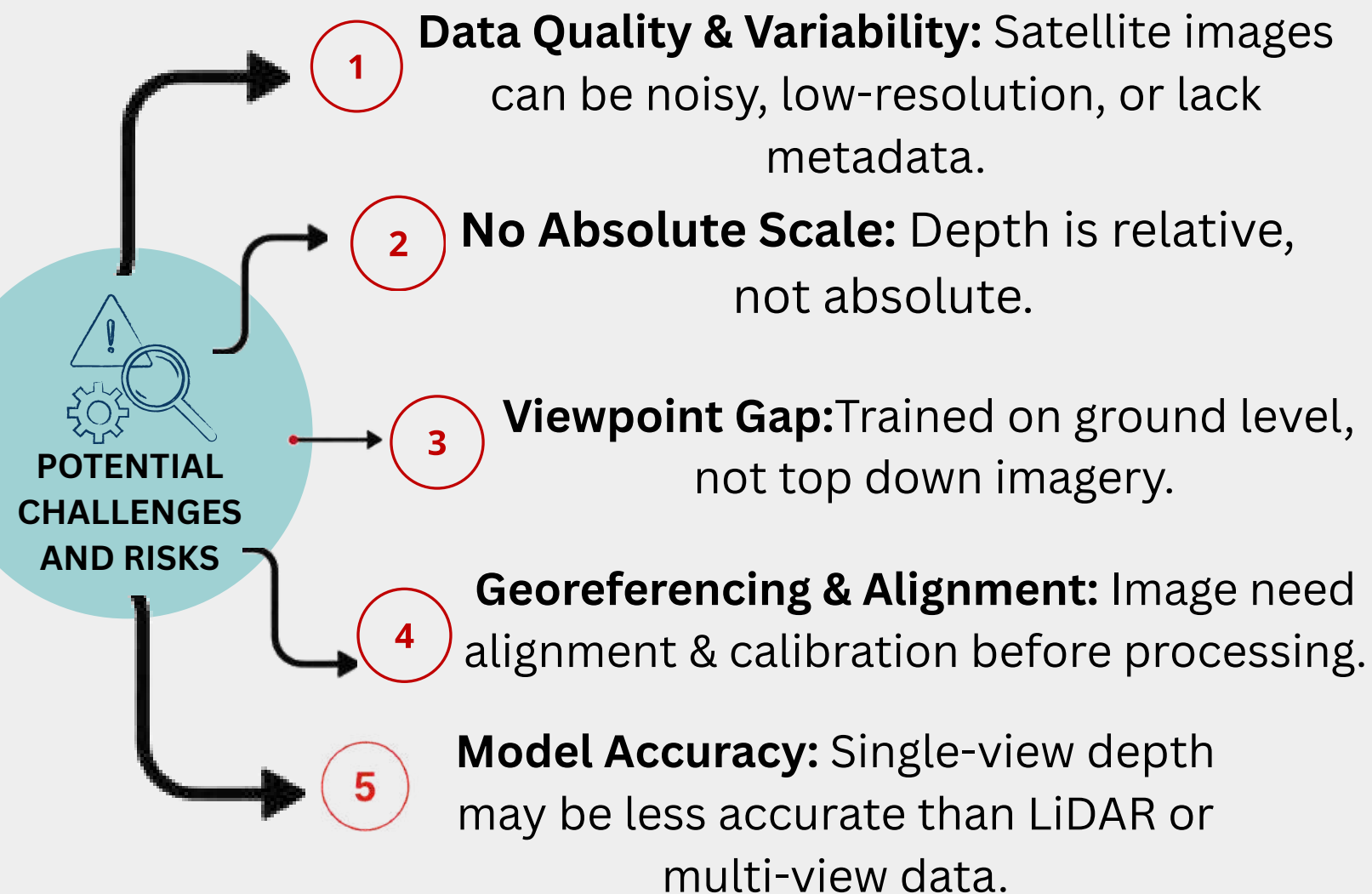
- **Technical** : Uses *publicly* available satellite/remote sensing images – no special hardware needed.
- **Economic** : Open-source tools (Python, OpenCV, GDAL, ensure **low cost** and **easy implementation**.
- **Data Feasibility** : **Uses existing datasets** (e.g. **GAMUS**, AHN4, swisstopo, COP30) – no need for new data collection.

VIABILITY

- **User-friendly interface** with simple upload and processing flow.
- Can be **integrated into existing geospatial platforms/web applications**.
- **Works anywhere on Earth** - and use cases(urban planning, disaster management,defence,etc).

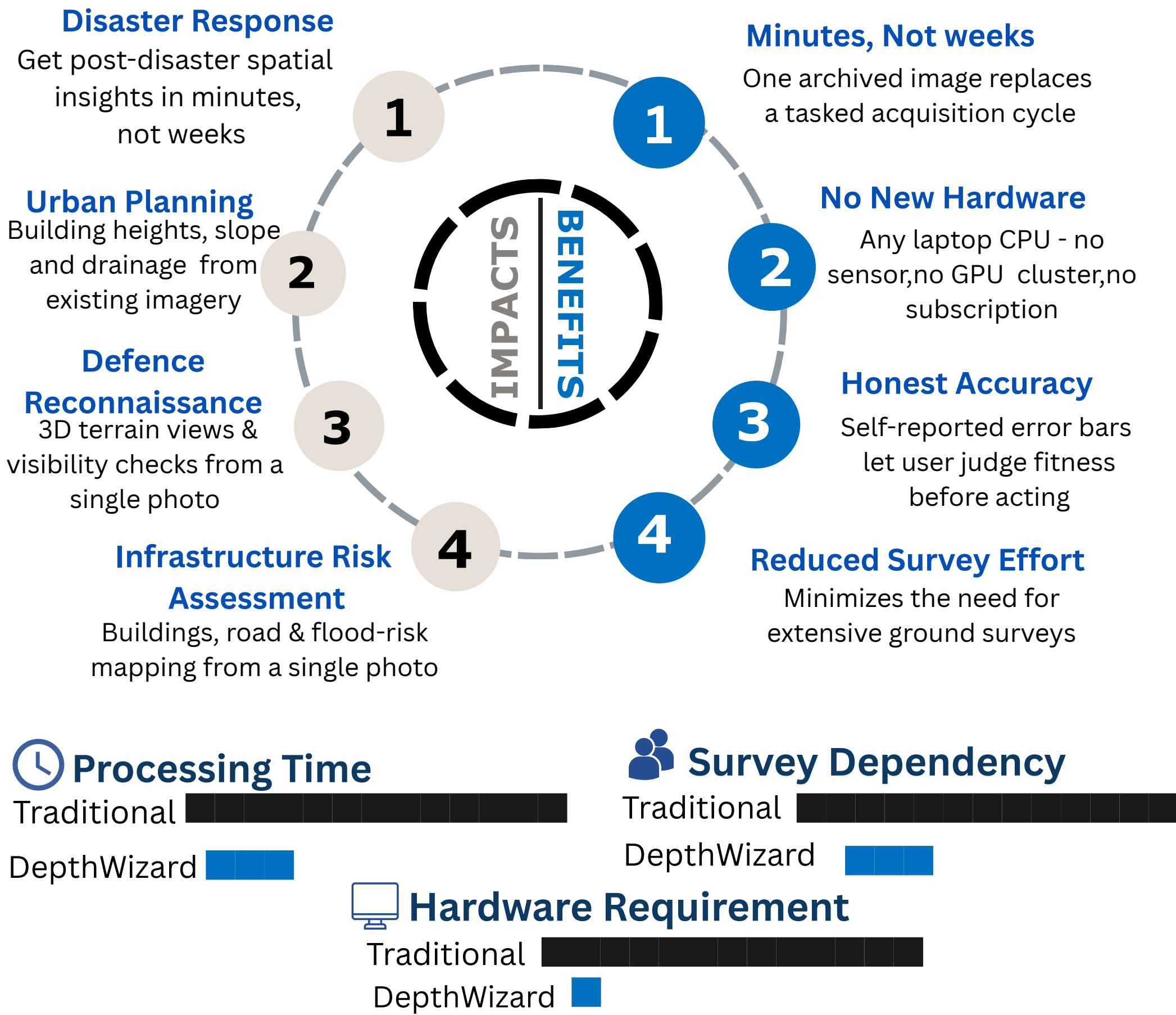
PRACTICAL IMPLEMENTATION

- **Market-ready** within **6-9 months** after prototype validation and testing.
- Works **offline** after initial data download (no constant internet required).
- Runs on a **laptop CPU** within half a minute.



STRATEGIES FOR OVERCOMING CHALLENGES

IMPACT AND BENEFITS



Commercialization & Market Opportunity

**India Geospatial Images Analytics Market**
(USD Million)

1,200

1,000

800

600

400

200

0

\$682.5M

\$1069.4M

2026

2030

11.9% CAGR (2026-2030)

**DepthWizard Revenue Systems**

**Institutional Licensing**

**Enterprise Deployment**

**Integration & Support**

Government & Civil Adoption Matrix

Government (Relevant Use Cases)	ADOPTION STAGE	Civil (Urban/Disaster)
ISRO Geospatial Mapping (urban planning, terrain & disaster)	Pilot	Urban Planning • Building heights • Slope & drainage • Master Planning
MoHUA GIS-based master planning and urban infrastructure		Disaster Management • Risk Assessment • Post-disaster analysis

RESEARCH PAPERS

1. Yang et al., "Depth Anything V2," NeurIPS 2024. : [LINK](#), project page : [LINK](#)
2. Mou & Zhu, "IM2HEIGHT : Height Estimation from Single Monocular Imagery," 2018. : [LINK](#)
3. He, Sun & Tang, "Guided Image Filtering," IEEE TPAMI 2013. : [LINK](#)
4. Xiong et al., "GAMUS: A Geometry-aware Multi-modal Semantic Segmentation Benchmark for Remote Sensing Data," arXiv 2023 : [LINK](#)
5. Chen et al., "HTC-DC Net: Monocular Height Estimation from Single Remote Sensing Images," IEEE TGRS 2023 : [LINK](#), arXiv : [LINK](#)

DATA SOURCES

1. GAMUS dataset (earthflow/GAMUS, CC-BY-4.0), used to fine-tune the depth model : [LINK](#)
2. Copernicus GLO-30 DEM via OpenTopography, used for the terrain baseline. : [LINK](#)
3. AHN4, the Dutch national LiDAR, used as reference for the Dutch scenes. : [LINK](#)
4. swisstopo swissSURFACE3D Raster, the Swiss LiDAR, used as reference for the Swiss scenes. : [LINK](#)
5. OpenAerialMap, Indian demo scene (CC-BY 4.0, credited) : [LINK](#)

OUR PROTOTYPE

